

Compact and Lightweight VR Glasses using Double Path Pancake Optics

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Abstract

We propose high light efficiency pancake optics for head mounted displays (HMDs) named “Double path (DP) pancake optics”, which achieve both compactness and high light efficiency. By analyzing and optimizing the lens design, we have dramatically improved the image quality through lens as evidenced by the modulation transfer function (MTF) in the Sagittal direction, which increases from nearly 0 to approximately 0.7 at spatial frequency of 12 line-pairs per mm (LP/mm) at a field of view (FOV) of 45 degrees. Furthermore, diopter adjustment from 0D to -9D can be realized, thanks to telecentric-like optical system.

Utilizing the improved DP pancake optics, we have launched compact and lightweight virtual reality (VR) glasses. The glasses weigh only 198g while employing an inside-out sensing method.

We believe that compactness and lightweight design are significant factor for user comfort, and the VR glasses equipped with DP pancake optics contribute to enhance the VR market.

Author Keywords

HMD; Compact HMD; Virtual reality; Polarized pancake optics; Folded optics; Diopter adjustment.

1. Introduction

The HMD market is expected to grow by approximately 30% year on year [1], and it is said that user comfort of wearing HMD is very important to expand the HMD market.

The folded HMD optics, commonly referred to as pancake optics, has been widely used in various products to achieve compact HMDs, such as Quest 3 by Meta, Galaxy XR by Samsung, and Apple Vision Pro by Apple. However, the conventional pancake optics has a significant drawback: Theoretical maximum light efficiency is only one-fourth (25%), due to light passing through a half mirror twice. To improve the light efficiency of pancake optics, we propose “DP pancake optics,” which utilizes two light paths. Our results indicate approximately 1.8 times higher light efficiency than conventional one [2]. Z. Luo et al. also propose DP pancake optics concept like ours [3]. They use cholesteric liquid crystal (CLC) instead of a combination of reflective polarizer (RP) and quarter wave plate (QWP). CLC can also perform the function of a concave mirror with a flat shape, commonly referred to as a polarization volume hologram (PVH). However, it may exhibit low contrast due to imperfection of the polarization state and show significant wavelength dependency depending on the incident angle. Y. Ding et al. also propose theoretically predicted 100% efficiency pancake optics using a Faraday rotator [4]. While this appears to be theoretically best approach, there are several issues that must be addressed for it to be applicable in a commercial HMD product, such as effective diameter, thickness, and cost, even if a commercial Faraday rotator can be used. Thus, we believe that, at present, DP pancake optics is the most promising technology to improve light efficiency in thin HMD optics like current pancake optics, and we are progressively refining our optical designs with the goal of mass production [5-7].

To achieve mass production, image quality through the lens must

be improved and the diopter adjustment should accommodate most users. In this paper, we introduce our approaches to addressing these issues, and present a commercial product named “Xrostella VR1”, which we have successfully launched.

2. Double Path Pancake Optics

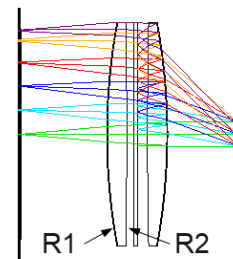
2.1. Advantage of Double Path Pancake

We propose our unique DP pancake optics realizing highly light efficiency, which has 1.8 times higher light efficiency than conventional one.

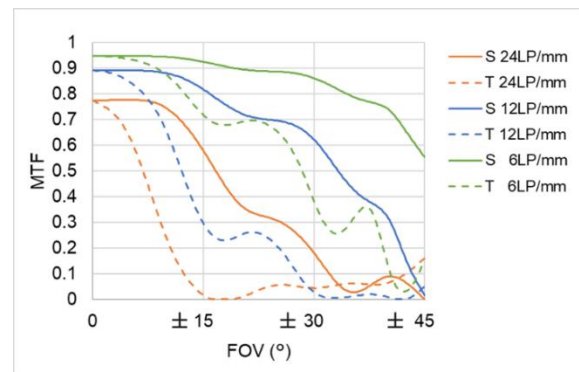
In the previous works, we established 2 basic optical designs, “Curved RP structure” (using 2lenses) and “Flat RP structure” (using 3 lenses) [2][5-7]. While Flat RP structure is higher productivity, Curved RP structure can achieve better image quality and lightweight. Therefore, we optimize the lens performance of Curved RP structure further by changing the optical design.

2.2. Design improvement for better MTF and to achieve diopter adjustment from 0 to -9D

Fig. 1 shows the ray diagram and the MTF chart of the previous design. The issue of the previous design is that MTF at high FOV region is insufficient while MTF near center region is good.



(a) Ray diagram



(b) MTF chart

Figure 1. Optical design of the previous design.

To address the issue, we evaluated the optical design by through-focus MTF (8.0 LP/mm). From this evaluation, we found focus points are slightly short at wide FOV region (The focus points are

above the display). Our DP pancake optics requires symmetric design, therefore, only R1 and R2 can be modified. Through this evaluation, we have found that restriction of magnification ratio to avoid screen-door-effect prevents better focus at wide FOV region. Therefore, we loosen the restriction of magnification ratio and found the optimized R2 lens design. This can improve through-focus MTF from approximately 1.1 mm above the display to within 0.4 mm above the display as shown in Fig. 2. Due to focus improvement, MTF at wide FOV region is improved as shown in Fig. 4.

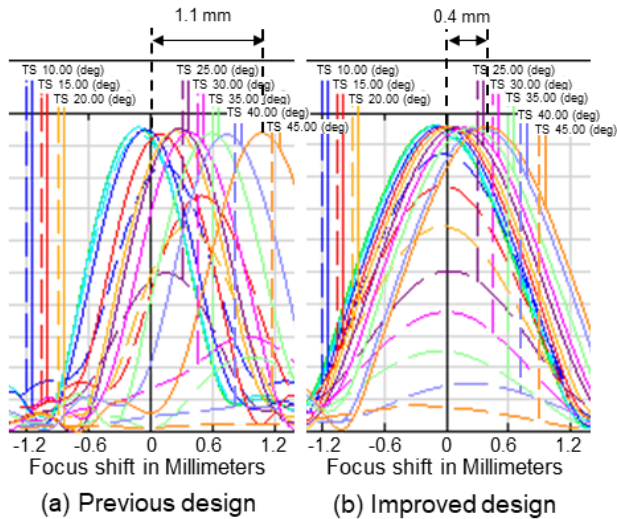


Figure 2. Through MTF chart.

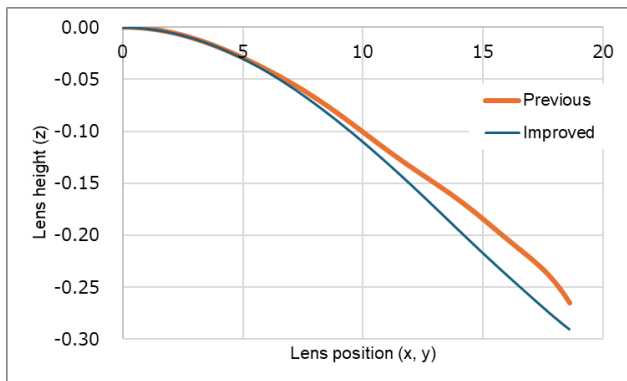


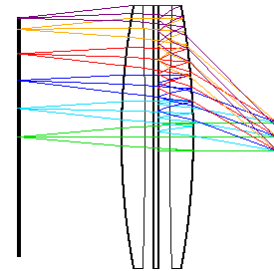
Figure 3. Optimized R2 lens design.

Our design concept prioritizes compactness and lightweight structures, which led us to incorporate mechanisms for interpupillary distance adjustment and diopter adjustment, thereby minimizing the size of the lenses as much as possible. Regarding diopter adjustment, competing products, such as the Meganex superlight 8K by Shiftall, offer compensation in the range of 0D to -7D [8]. However, it is recognized that to accommodate a wider range of vision impairments, adjustments from 0D to -9D are necessary. In response, we reevaluated the lens design and discovered that we could achieve this compensation by adopting an optical system that closely resembles a one-sided telecentric configuration.

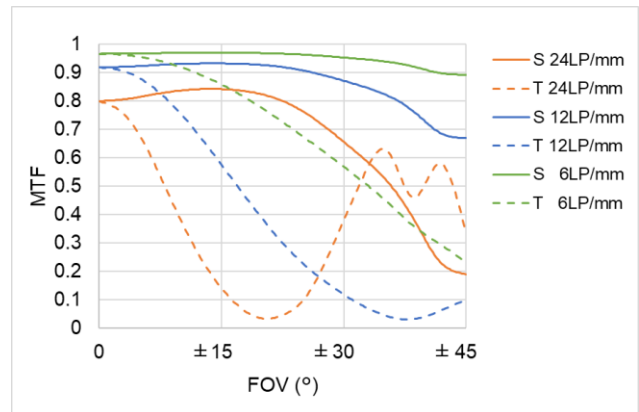
A one-sided telecentric optical system refers to a configuration in

which the principal rays are parallel to the optical axis at the object plane (or the image plane), as illustrated in Fig. 5. Here, it specifically denotes the optical system in which light parallel to the optical axis emitted from the display enters the human eye (pupil).

Thanks to the telecentric-like optics, magnification ratio from 0D to -9D is not so changed as shown in Fig. 6. The magnification ratio of center region $\pm 5^\circ$ is almost constant and, even in $\pm 40^\circ$ region, magnification ratio is from 13 to 14.



(a) Ray diagram



(b) MTF chart

Figure 4. Optical design of the improved design.

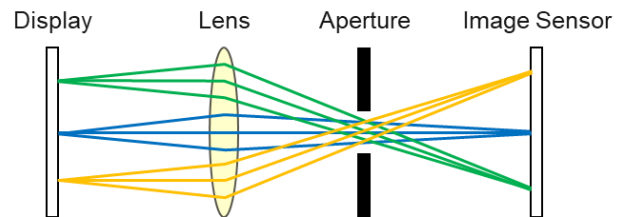


Figure 5. One-sided telecentric optics.

This optical system enables to achieve compensation up to -9D while maintaining excellent image quality, as shown in Fig. 7, even when the diopter adjustment is altered, with only minimal changes in magnification.

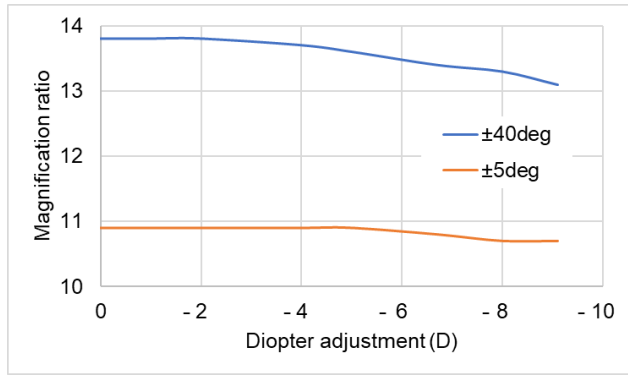
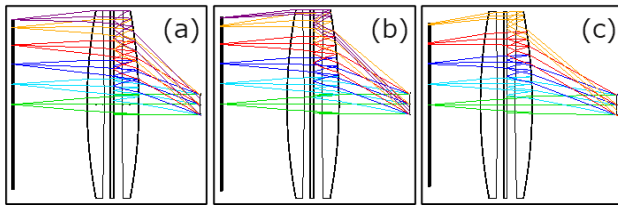
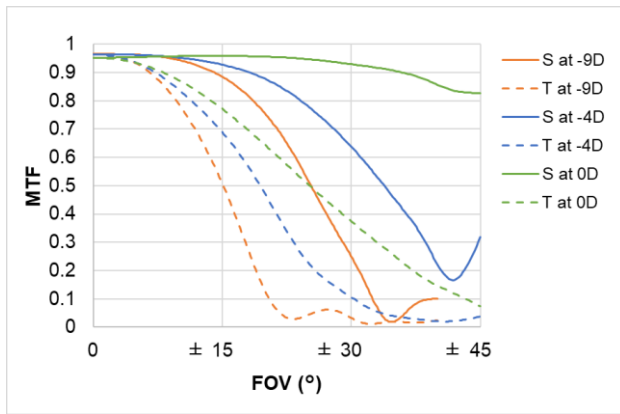


Figure 6. Magnification ratio from 0D to -9D.



Ray diagram at (a) 0D, (b) -4D, and (c) -9D



(b) MTF at 8LP/mm

Figure 7. Optical design at 0D, -4D, and -9D.

2.3. Prototype with 90 deg FOV

According to the design mentioned above, we have fabricated the prototype with 90 deg FOV. Fig. 8 shows the appearance of the prototype. Additionally, Table 1 shows specification and Table 2 shows transmittance of the prototype. To compare with the commercial pancake optics we can get, 1.8 times light efficiency can be obtained as expected.



(a) Top view (b) Side view
Figure 8. Appearance of 90 deg FOV prototype.

Table 1. Specifications of prototype.

	Items	Specifications
LCD	Active area size	Approx. 2.5 inch
	Resolution	Approx. 1,200 ppi
Optics	Thickness of optics	25.0 mm
	Weight	28.0 g
	FOV	90 deg
	Lens diameter	34 mm
	Efficiency	1.8 times
	Magnification	Approx. 15
	Eye relief	12.0 mm
	Focal length	23.16 mm

Table 2. Measurement result of total light transmittance

	Double path	Single path
Total light transmittance	42.2%	23.1%
Transmittance ratio	1.8	1

2.4. Measurement result of MTF

After fabricating the prototype, the MTF were measured using our simple evaluation system as shown in Fig. 9. Our system utilizes a display as the source of the MTF charts, and the MTF charts displayed have been captured through the optics by a camera with an aperture of approximately 3mm in diameter. While these MTF values are not so strictly precise, we believe they are useful for comparing the performance of the optics. We also have evaluated a commercially available pancake optics as a reference (Ref). The improved prototype demonstrates better MTF performance than both the previous version and the Ref.

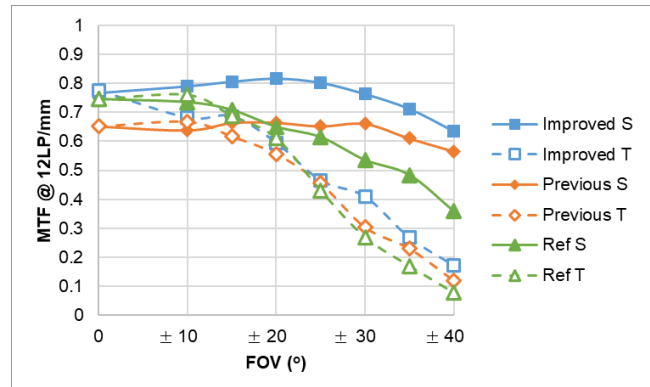


Figure 9. Measured MTF.

3. Features of our VR Glasses

We have fabricated a novel compact and lightweight HMD product utilizing our DP pancake optics mentioned above. Fig. 9 shows the appearance of the product, and Table 3 describes the main specifications of the product. Lightweight can realize glasses-like appearance. Additionally, by incorporating foldable temples, it is easier for users to attach and detach the VR housing. Considering a case with intense head movement, the band to wrap around the back of the head for added stability is also prepared.

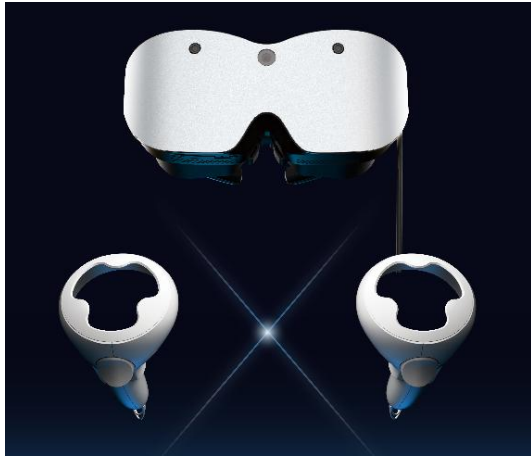


Figure 10. Compact VR Glasses “Xrostella VR1.”

Table 3. Specifications of “Xrostella VR1”

Items	Specifications
Weight	198g
Optics	Double path pancake optics (= 1.8 times light efficiency)
Eyesight correction	From 0D to -9D
Sensing method	Inside-out

We adopt PC and Smartphone connected model to achieve lightweight. This model can remove batteries and several chipsets which are burden of weight.

There are generally two approaches to self-position estimation: the Inside-out method and the Outside-in method. The Inside-out method requires the incorporation of cameras and inertial sensors, in contrast to the Outside-in method. Additionally, it necessitates that image processing and self-position estimation be performed on the headset side, which requires a chipset, potentially making the HMD housing heavier. However, in our pursuit of enhanced usability, we intentionally aimed for a lightweight design while employing the Inside-out method.

4. Conclusion

This paper reports on the improvements we have made to the high-efficiency pancake lens (double-path method), revisiting the lens design to achieve diopter adjustment from 0D to -9D while maintaining high MTF performance. With this level of MTF, we believe the lens has significant market potential. Although we altered the lens shape, we achieved an optical utilization efficiency of approximately 1.8 times, contributing to lower power consumption.

In an all-in-one model, this power consumption reduction contributes to smaller battery sizes, thus aiding in weight reduction. However, even in PCs or smartphones connected model, it can contribute to mitigate temperature increases caused by heat generation. Since VR HMDs are worn on the face and contain displays, heat emitted from the displays can elevate the internal

temperature of the headset, leading to user discomfort. Our design can help prevent this issue.

5. Impact of the technology

Our DP pancake optics design, which can reduce the power consumption and heat generation. Currently, this method has been applied with a wired connection model (with PCs or Smartphones); however, we believe it can also be adapted for an all-in-one model in the future, contributing to reduced battery size and addressing heat generation issues. We hope that this high optical utilization efficiency lens technology, combined with our lightweight VR glasses, “Xrostella,” will contribute to the expansion of the VR HMD market.

6. Acknowledgements

The authors thank connected solution business unit in SHARP Corporation and all who related for their help in making the HMD headset prototype successfully using the double path pancake optics prototype mentioned above.

7. References

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